

A Data & Programming Approach to Predicting Key Drivers of Airline Customer Satisfaction

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1. Introduction and Project Overview

This project analyzes passenger records from the *Airline Quality Ratings* dataset to identify which aspects of the flight experience most strongly influence satisfaction. After preparing the data and conducting exploratory analysis, we applied logistic regression to predict satisfaction and quantify the impact of each service variable. The final model achieved a strong accuracy, demonstrating the ability to distinguish satisfied from dissatisfied customers. Our analysis highlights several powerful satisfaction drivers, such as online boarding, in-flight Wi-Fi, seat comfort, cleanliness, and leg room, and identifies key pain points, including long delays during arrival, economy seating, and personal leisure travel. Based on these insights, we propose strategic recommendations that airlines can translate directly into operational improvements: enhancing digital convenience, targeting comfort upgrades for economy passengers, improving arrival-time reliability, and reinforcing loyalty among returning customers.

1.1 Background and Motivation for the Study

Air travel is a major part of many people's lives, and airlines depend on customer satisfaction to stay competitive and build loyalty. According to the American Customer Satisfaction Index, in recent years, industry reports have shown a drop in passenger satisfaction, with issues like delays, limited comfort, and inconsistent service (2025). Since airlines collect detailed ratings on travel experiences (ACSI, 2025), data analysis can help identify which factors matter most to passengers. By studying these ratings, we can understand what actually drives satisfaction and where airlines should focus their improvements.

Given the recent decline in airline satisfaction and the wide range of factors that influence the travel experience, an important question arises: "What specific travel experience factors should airlines prioritize to improve passenger satisfaction and strengthen customer loyalty?"

2. Related Work / Relevant Studies

According to a 2022 study titled *Investigating Airline Passenger Satisfaction: Data Mining Approach*, the strongest drivers of passenger satisfaction are factors related to comfort, in-flight service quality, and overall reliability (Noviantoro & Huang). The study found that elements such as seat comfort, service interactions, and timeliness significantly influence how passengers rate their travel experience (Noviantoro & Huang, 2022). These findings support the

idea that satisfaction is shaped by both service quality and operational performance, which aligns closely with the variables in our dataset. The study also shows the value of analyzing detailed, feature-level passenger data, an approach we follow in this project to better understand which specific travel experience factors matter most.

3. Data

3.1 Data Source (Kaggle Airline Quality Ratings)

The source of our data is the *Airline Quality Ratings* dataset available on Kaggle, a publicly available source. It contains 129,880 passenger records and 24 features, which include demographics, travel information, time metrics, multiple service quality ratings, and satisfaction level. This dataset provides a thorough, structured view of customer experience.

airline.describe()										
	ID	Age	Flight Distance	Departure Delay	Arrival Delay	Departure and Arrival Time Convenience	Ease of Online Booking	Check-in Service	Online Boarding	Gate Location
count	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000
mean	64940.500000	39.427957	1190.316392	14.713713	15.091129	3.057599	2.756876	3.306267	3.252633	2.97
std	37493.270818	15.119360	997.452477	38.071126	38.465650	1.526741	1.401740	1.266185	1.350719	1.27
min	1.000000	7.000000	31.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00
25%	32470.750000	27.000000	414.000000	0.000000	0.000000	2.000000	2.000000	3.000000	2.000000	2.00
50%	64940.500000	40.000000	844.000000	0.000000	0.000000	3.000000	3.000000	3.000000	3.000000	3.00
75%	97410.250000	51.000000	1744.000000	12.000000	13.000000	4.000000	4.000000	4.000000	4.000000	4.00
max	129880.000000	85.000000	4983.000000	1592.000000	1584.000000	5.000000	5.000000	5.000000	5.000000	5.00

3.2 Feature Descriptions

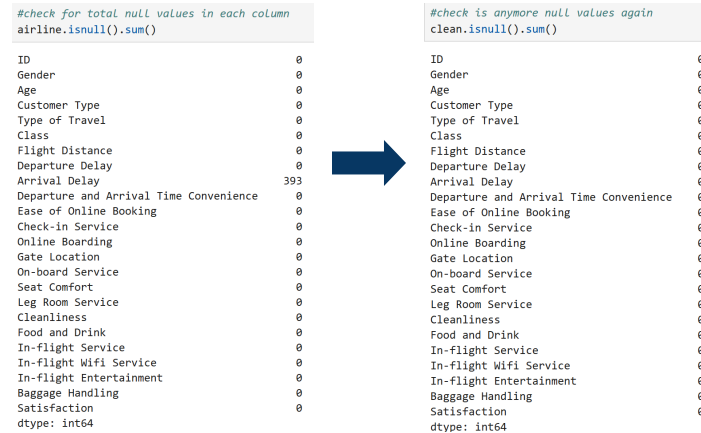
The dataset includes four main categories of variables. Demographic variables describe passenger characteristics such as age, gender, and customer type. Travel-related variables include type of travel, travel class, and flight distance. Operational metrics capture delay information, including departure delay and arrival delay. Service quality variables (i.e., seat comfort, cleanliness, online boarding, in-flight wifi, in-flight service, etc.) additionally provide ratings on a scale of 1-5. Lastly, the satisfaction variable serves as the study's target variable and contains the values "Satisfied" and "Neutral or Dissatisfied".

4. Data Preparation

4.1 Data Cleaning

To ensure the dataset was suitable for analysis and modeling, we performed several data cleaning procedures. The first step involved identifying missing values across all variables. As

shown in the output, only the arrival delay column contained missing entries, totaling 393 observations. Since these missing values represented less than 0.3 % of the dataset and were isolated to one numeric variable, we chose to remove those rows rather than impute them. With this approach, we preserved data integrity without affecting the representativeness of the sample. After removal, a second check confirmed that the dataset no longer contained missing values.



#check for total null values in each column airline.isnull().sum()		#check is anymore null values again clean.isnull().sum()	
ID	0	ID	0
Gender	0	Gender	0
Age	0	Age	0
Customer Type	0	Customer Type	0
Type of Travel	0	Type of Travel	0
Class	0	Class	0
Flight Distance	0	Flight Distance	0
Departure Delay	0	Departure Delay	0
Arrival Delay	393	Arrival Delay	0
Departure and Arrival Time Convenience	0	Departure and Arrival Time Convenience	0
Ease of Online Booking	0	Ease of Online Booking	0
Check-in Service	0	Check-in Service	0
Online Boarding	0	Online Boarding	0
Gate Location	0	Gate Location	0
On-board Service	0	On-board Service	0
Seat Comfort	0	Seat Comfort	0
Leg Room Service	0	Leg Room Service	0
Cleanliness	0	Cleanliness	0
Food and Drink	0	Food and Drink	0
In-flight Service	0	In-flight Service	0
In-flight Wifi Service	0	In-flight Wifi Service	0
In-flight Entertainment	0	In-flight Entertainment	0
Baggage Handling	0	Baggage Handling	0
Satisfaction	0	Satisfaction	0
dtype: int64		dtype: int64	

In addition to handling missing values, we verified the consistency of variable types, ensuring variables were free from formatting errors.

4.2 Feature Encoding

For categorical variables such as Gender, Customer Type, Type of Travel, and Class, the One Hot Encoding method was deployed to convert them into binary variables that could be interpreted by the model. As mentioned earlier, the Satisfaction variable was encoded into a binary format as well, allowing us to use it as a response variable for our model.

For modeling purposes, the Satisfaction variable was converted into a binary target, where 1 was mapped to “Satisfied” and 0 was mapped to “Neutral or Dissatisfied.” The figure below indicates the distribution of satisfaction categories in the cleaned dataset.

```
Hit Count:
Satisfaction
Neutral or Dissatisfied    73225
Satisfied                  56262
Name: count, dtype: int64
Hit Proportion:
Satisfaction
Neutral or Dissatisfied    0.565501
Satisfied                  0.434499
Name: proportion, dtype: float64
```

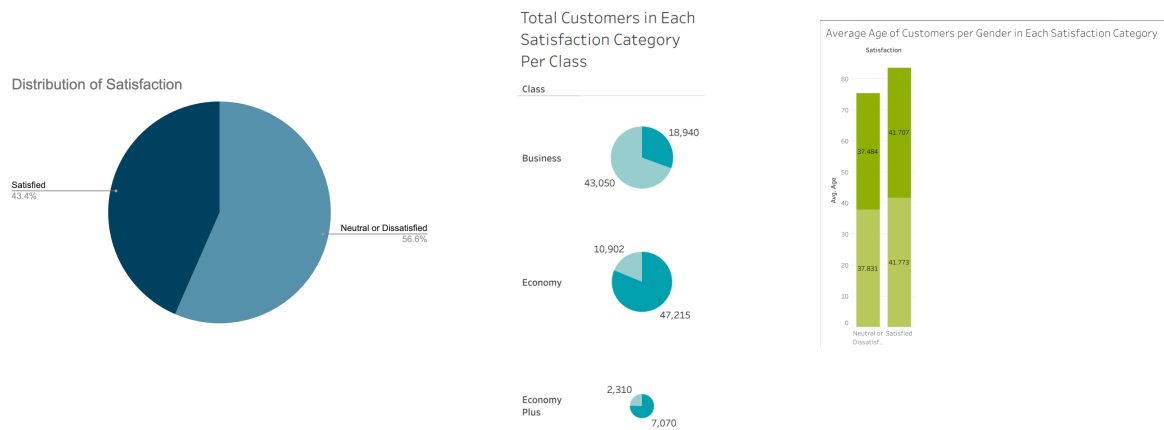
Weights were considered unnecessary to apply in the model due to the insignificant difference in proportions between satisfaction categories (57:43).

4.4 Final Dataset Used for Analysis

The updated count of passenger reviews after cleaning is 129,487, as indicated in the figure below.

	ID	Age	Flight Distance	Departure Delay	Arrival Delay	Departure and Arrival Time Convenience	Ease of Online Booking	Check-in Service	Online Boarding	Gate Location	On-board Service	Seat Comfort	Leg Room Service	Cleanliness	Food and Drink	In-flight Service	In-flight Wifi Service	In-flight Entertainment	Baggage Handling
count	129487.000000	129487.000000	129487.000000	129487.000000	129487.000000	129487.000000	129487.000000	129487.000000	129487.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000
mean	64958.335169	39.428761	1190.210662	14.643385	15.091129	3.057349	2.756786	3.306239	3.252720	2.976925	3.383023	3.441361	3.350878	3.286326	3.204774	3.642193	2.728696	3.358077	3.632114
std	37489.781165	15.117597	997.560954	37.932867	38.465650	1.526787	1.401662	1.266146	1.350651	1.278520	1.287099	1.319289	1.316252	1.313682	1.329933	1.176669	1.329340	1.334049	1.180025
min	1.000000	7.000000	31.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000
25%	32494.500000	27.000000	414.000000	0.000000	0.000000	2.000000	2.000000	3.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	3.000000	2.000000	2.000000	3.000000
50%	64972.000000	40.000000	844.000000	0.000000	0.000000	3.000000	3.000000	3.000000	3.000000	3.000000	4.000000	4.000000	4.000000	4.000000	4.000000	4.000000	3.000000	4.000000	4.000000
75%	97415.500000	51.000000	1744.000000	12.000000	13.000000	4.000000	4.000000	4.000000	4.000000	4.000000	4.000000	5.000000	4.000000	4.000000	4.000000	5.000000	4.000000	4.000000	5.000000
max	129880.000000	85.000000	4983.000000	1592.000000	1584.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000

5. Exploratory Data Analysis (EDA)



As shown above, the first pie chart visualizes the distribution of each satisfaction category in the dataset, showing that there is a slightly higher percentage of “Neutral or Dissatisfied” reviews (56.5%) than “Satisfied reviews (43.4%) in the dataset. The second pie chart goes into greater detail, separating this distribution by Plane Class. Within the business class, there seems to be a significantly higher proportion of “Satisfied” reviews (~67%) compared to “Neutral or Dissatisfied” reviews, while in the other two classes, the proportion of “Neutral or Dissatisfied” reviews is much higher.

Lastly, the bar chart demonstrates the average age of customers per gender in each satisfaction category. The mean age of both female and male customers who left “Neutral or Dissatisfied” reviews was around 37, while in the “Satisfied” category, the average age of both genders was around 41. The age difference between satisfaction groups is roughly 4 years, indicating that older passengers tend to report higher satisfaction levels. In contrast, gender

differences within each satisfaction category are minimal, suggesting that age plays a more meaningful role than gender in explaining satisfaction differences. It is important to note that all visuals displayed in this section were developed using Tableau.

6. Methodology

6.1 Modeling Approach Overview

To identify which travel experience factors best predict passenger satisfaction, we used a logistic regression model suited for our binary outcome of “Satisfied” (1) and “Neutral or Dissatisfied” (0). The analysis pipeline followed a structured progression: rigorous data cleaning, feature encoding, variable standardization, exploratory insights, model construction, and post-model diagnostics. This created a clear analytical audit trail and ensured the final results reflected reliable patterns.

6.2 Why Logistic Regression Was Selected

Logistic regression was selected because it naturally fits our binary outcome variables (“Neutral or Dissatisfied” vs. “Satisfied”) and provides clear, interpretable insights into how each factor may influence passenger satisfaction level (GeeksforGeeks, 2017). The model accommodates both categorical and numerical variables with minimal preprocessing, making it well-suited for this dataset. In addition, logistic regression provides a transparent and reliable baseline for evaluating the impact of individual service and experience factors on satisfaction.

6.3 Model Assumptions & Interpretability

There are key assumptions typical of logistic regression that must be considered. Earlier, our dependent variable was converted into a binary format, which is a fundamental requirement for logistic regression. The model also assumes a linear relationship between predictor variables and the log odds of Satisfaction, a reasonable approximation given the numerical ratings for service quality and the categorical encodings used. Observations are also treated independently, which is appropriate considering each passenger represents a distinct travel experience. Multicollinearity was examined indirectly through distribution summaries and the stability of coefficient estimates. (GeeksforGeeks, 2017).

Due to the large number of observations in the sample, the coefficient estimates are also ensured to be robust. Python libraries such as statsmodels made interpretability central to the analysis by producing coefficient tables and enabling downstream odds-ratio calculation; meanwhile, libraries such as sklearn provided a method to evaluate model performance metrics.

6.4 Model Setup

After cleaning the dataset and encoding the categorical variables using the one-hot encoding method, the dataset was prepared to be used for modeling. The data was divided into training and test sets using an 80:20 ratio, which can be interpreted as 80% of the data being used to train the model, while 20% of the remaining data was used to evaluate the model's performance. The training accuracy was 83.45%, which demonstrated good performance on the data it was trained on, and the test accuracy was 82.96%, also indicating quality performance on data that the model was not trained on.

Training accuracy: 0.8345094556371816
Test accuracy: 0.82960074136999

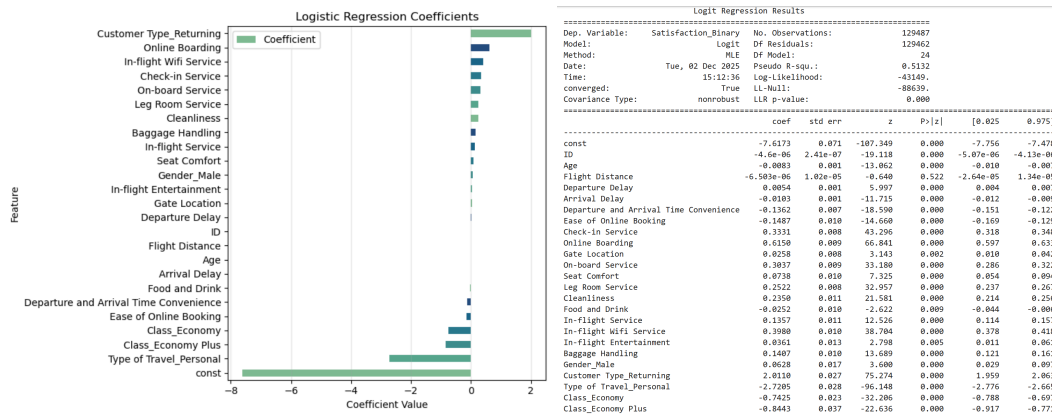
Furthermore, the classification report provides evaluation metrics for model precision, recall, f1-score, and accuracy. For neutral or dissatisfied reviews (class 0), the model has a precision score of 0.86 (predicts "Neutral or Dissatisfied" correct 86% of the time), a recall score of 0.83 (identifies only 83% of actual neutral/dissatisfied customers), and an F1-score of 85% (great at balancing precision and recall). Meanwhile, when studying satisfied reviews (class 1), the model has a precision score of 0.79 (predicts "Satisfied" correct 79% of the time), a recall score of 0.82 (identifies only 82% of actual neutral/dissatisfied customers), and an F1-Score of 81% (great at balancing precision and recall). In regard to overall accuracy, the model had a score of 83%, which demonstrates high performance quality, as shown in the image below.

Classification Report:				
	precision	recall	f1-score	support
0	0.86	0.83	0.85	14685
1	0.79	0.82	0.81	11213
accuracy			0.83	25898
macro avg	0.83	0.83	0.83	25898
weighted avg	0.83	0.83	0.83	25898

7. Results

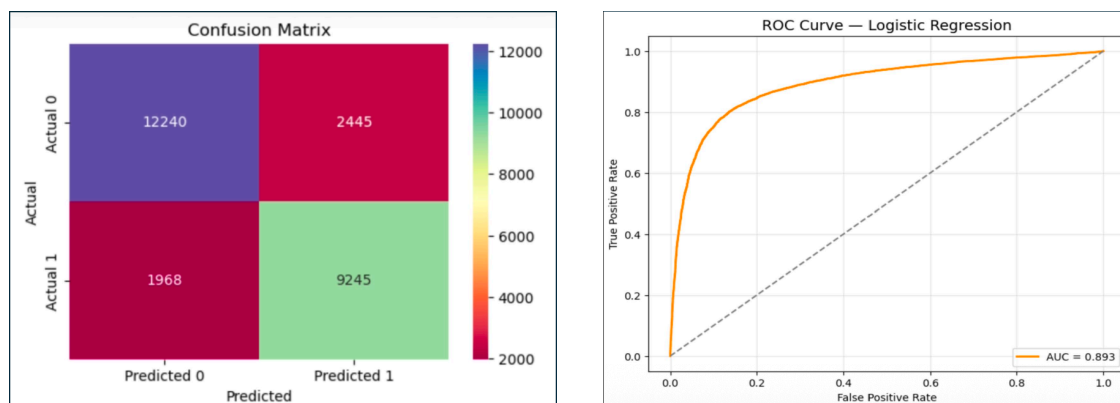
7.1 Logistic Regression Coefficient Interpretation

The logistic regression results revealed that returning customers and comfort-related features, such as online boarding, reliable Wi-Fi, seat comfort, leg room, and cleanliness, were the strongest drivers of satisfaction. In contrast, economy-class travel, personal travel, and especially arrival delays lowered satisfaction, reflecting well-known passenger pain points. Overall, the findings show that people value a smooth, comfortable, and timely journey, and satisfaction declines sharply when those expectations aren't met.



7.2 Further Model Performance Evaluation

The logistic regression model achieves solid predictive accuracy across both classes. The confusion matrix shows that the model correctly identified 12,240 neutral/dissatisfied passengers and 9,245 satisfied passengers, while misclassifying 2,445 neutral/dissatisfied and 1,968 satisfied passengers. Overall, these results indicate balanced performance and reliable separation between the two groups. The ROC curve further supports this, with an AUC of 0.893, showing strong discriminative power and reliable separation between satisfied and unsatisfied passengers.



8. Managerial Implications

8.1 Which Factors Airlines Should Prioritize

Our analysis shows that some service features have a much stronger impact on passenger satisfaction than others. The biggest positive drivers are online boarding, seat comfort, leg room, in-flight Wi-Fi, cleanliness, and on-board service. These features consistently received higher ratings from satisfied passengers and also showed strong positive logistic regression coefficients. Negative predictors also matter. Attributes such as delays, travel class, and customer type significantly reduce satisfaction when conditions are poor. Airlines should focus most on improving comfort, convenience, and reliability, since these factors shape how passengers feel about the entire flight.

8.2 How Airlines Can Use These Insights

Airlines can use these insights to make better decisions about where to invest time and resources. Improving high-impact areas like seating, Wi-Fi quality, cleanliness, and boarding processes can boost satisfaction with manageable changes. The finding that returning customers tend to be more satisfied also suggests that loyalty programs and consistent service can increase long-term customer value. Meanwhile, negative predictors highlight where operational problems occur. For example, since delays strongly decrease satisfaction, airlines can review scheduling, staffing, or maintenance processes that cause delays.

9. Recommendations

Based on our insights, we developed four targeted recommendations to improve passenger satisfaction. First, we propose the Returners Club Lite initiative, which offers simple perks such as early seat selection and free Wi-Fi vouchers to strengthen satisfaction among returning passengers, who emerged as the strongest positive indicator in our model. Second, the Comfort Micro-Upgrades Program focuses on small but high-impact improvements, including memory-foam seat pads, improved footrests, and clearer cleanliness certification, all of which enhance perceived comfort without requiring major aircraft modifications. Third, we recommend a Smart Boarding Experience that reduces digital friction through features like one-tap check-in and Wi-Fi reliability indicators, directly addressing convenience factors that strongly influence satisfaction. Finally, the Arrival Precision Program shifts operational focus toward on-time

arrivals by improving gate allocation, arrival-time prediction, and ground coordination, since late arrivals were shown to reduce satisfaction far more than late departures.

10. Limitations

The dataset represents one period of data collection, so it may not capture seasonal or long-term changes in customer expectations. Satisfaction ratings are self-reported, which introduces personal bias. Logistic regression assumes a linear relationship between predictors and the log-odds of satisfaction, which can cause oversimplification of more complex interactions. The model also cannot account for influential predictors that are not included in the dataset such as ticket price, staff behavior, aircraft age, noise levels, airport experience, or exact seat type are not included in the dataset. These findings must be interpreted as helpful guidance rather than universal rules.

11. Conclusion

This project examined which aspects of the airline travel experience have the strongest impact on passenger satisfaction. Using exploratory analysis and a logistic regression model, the most influential features were identified. Together, these results provide a clear view of which service components matter most to passengers. The value of this analysis lies in its ability to translate a large, complex dataset into actionable insights for airlines. Instead of assuming which service elements matter, the model identifies them using data-driven evidence. Airlines can use these insights to prioritize improvements that will have the greatest impact on customer experience, loyalty, and long-term profitability. The combination of EDA and logistic regression also demonstrates how analytical tools can support strategic decision-making in industries where customer satisfaction directly influences business outcomes.

In the future, other modeling approaches could be compared with logistic regression to explore potential nonlinear relationships. Collecting data across different airlines, seasons, or regions would help improve generalizability and provide a broader understanding of passenger preferences over time.

12. Bibliography

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Slides Presentation

UP IN THE AIR: PREDICTING PASSENGER SATISFACTION

~what really makes a flight "worth" it!

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AGENDA



01. Problem Statement
02. Business Implications
03. Data Overview
04. Data Preparation
05. Exploratory Analysis
06. Logistic Regression
07. Analysis
08. Insights
09. Recommendations
10. Limitations

Problem Statement

What specific travel experience factors should airlines prioritize to improve passenger satisfaction and strengthen customer loyalty?



Business Implications



- 1 Higher satisfaction → Stronger customer loyalty
↑ Repeat Bookings ↑ Airlines Profitability
- 2 Targeted Improvement
Airlines focus resources on factors that statistically matter
- 3 Competitive in the Market

Data Overview

129,880

Passenger
observations

24

Features

10.0

Kaggle
usability score



Data Preparation



Removed missing
and inconsistent
values



Converted
satisfaction into a
binary target
variable



Encoded
categorical
variables (gender,
class, travel type,
customer type)



Standardized
numerical
variables (age,
distance, delays)



Evaluated feature
correlations and
distributions



80:20 train:test

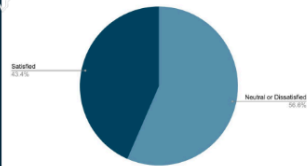


Checked
distribution of
dependent
variable
56:43 distribution
→ no weights

Exploratory Analysis



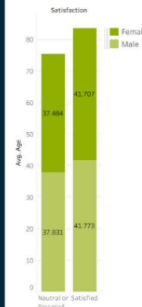
Distribution of Satisfaction



Interesting Insights from Data

- Avg age: 39 | Range: 7 - 85
- Delay (median = 0), but both departure and arrival have extreme outliers (> 1,500 mins)
- Most rating-based service variables cluster between 3-4

Average Age of Customers per Gender in Each Satisfaction Category



Total Customers in Each Satisfaction Category Per Class



Predictive Modeling

? We chose Logistic Regression

- Target variable is binary (0 = "Neutral or Dissatisfied", 1 = satisfied)
- Produces interpretable coefficients & odds ratios
- Useful for understanding which features matter most
- Handles both numerical and categorical features effectively

Model Evaluation

Training accuracy: 0.8345094556371816
Test accuracy: 0.82960074136999

Confusion Matrix:
[[12240 2445]
 [1968 9245]]

Classification Report:				
	precision	recall	f1-score	support
0	0.86	0.83	0.85	14685
1	0.79	0.82	0.81	11213
accuracy			0.83	25898
macro avg	0.83	0.83	0.83	25898
weighted avg	0.83	0.83	0.83	25898

→ Accuracy: ~83%

→ Neutral or Dissatisfied (class 0):

- predicts "Neutral or Dissatisfied" correct 86% of the time
- identifies only 83% of actual neutral/dissatisfied customers
- F1: 85%

→ Satisfied(class 1):

- predicts "satisfied" correct 79% of the time
- identifies 82% of actual satisfied customers
- F1: 81%

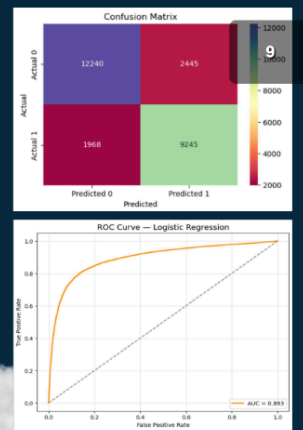
Model Evaluation

Confusion Matrix:

- 12240 of Neutral or Dissatisfied customers → predicted "Neutral or Dissatisfied"
- 2445 of Neutral or Dissatisfied customers → predicted "Satisfied"
- 1968 of Satisfied customers → predicted "Neutral or Dissatisfied"
- 9245 of Satisfied customers → predicted "Satisfied"

ROC Curve:

- AUC = 0.893



High Positive Coefficients:

- Returning Customers, Online Boarding, Wi-Fi, Seat Comfort, Cleanliness, Leg Room

Negative Coefficients:

- Economy class, personal travel, long delays

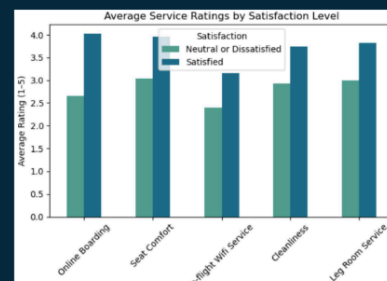


Analysis

Odds Ratio:	Feature	Coefficient	Odds Ratio
0	Customer Type_Returning	2.011045	7.471118
1	Online Boarding	0.514981	1.669521
2	In-flight Wi-Fi Service	0.397982	1.488817
3	Check-in Service	0.333112	1.395303
4	On-board Service	0.303743	1.354921
5	Leg Room Service	0.252190	1.286841
6	Cleanliness	0.235083	1.264912
7	Baggage Handling	0.140656	1.151029
8	In-flight Service	0.135726	1.145367
9	Seat Comfort	0.073828	1.076621
10	Gender_Male	0.062828	1.064843
11	In-flight Entertainment	0.036058	1.036716
12	Gate Location	0.025768	1.026103
13	Departure Delay	0.005354	1.005369
14	ID	-0.000005	0.999995
15	Flight Distance	-0.000007	0.999993
16	Age	-0.008344	0.991691
17	Arrival Delay	-0.010335	0.989718
18	Food and Drink	-0.025186	0.975128
19	Departure and Arrival Time Convenience	-0.136241	0.872632
20	Ease of Online Booking	-0.148691	0.861836
21	Class_Economy	-0.742501	0.475922
22	Class_Economy Plus	-0.844312	0.429853
23	Type of Travel_Personal	-2.720478	0.065843
24	const	-7.617290	0.000492

Analysis

- Satisfied passengers
 - higher service ratings
- Biggest rating gaps
 - seat comfort, online boarding
- Early indication that comfort + convenience
 - strongly influence satisfaction



Insights

- Convenience features (online boarding, Wi-Fi) → strongest satisfaction drivers
- Comfort factors (seat comfort, leg room, cleanliness) show biggest rating gaps
- Returning passengers are much more satisfied
- Economy class + long delays strongly reduce satisfaction
- Passengers care more about when they land > leave
- Model achieves ~83% accuracy → results are reliable and actionable



Recommendations

1



Returners Club Lite

- Fast lane lite during off-peak season
- Free WiFi vouchers & early seat selection
- Reward customers for consecutive flights

2



Comfort Micro-Upgrades Program

- Memory-foam seat pads for economy
- Adjusted footrests in long-haul seats for economy
- Sanitation cards on seats

3



Smart Boarding Experience

- Smart online boarding pass
- 1-Tap mobile check-in
- Wi-Fi guarantee Badge

4



Arrival Precision Program

- Coordinated descent optimization
- "Last Mile Prediction" Alert
- Gate Stability algorithm

Limitations

- Missing real-world operational context
- Self-reported ratings may be inconsistent/bias
- Binary satisfaction (0 or 1) oversimplifies passenger sentiment
- Highly related service variables may overlap in effect
- Lack of airline-specific effects



THANK
YOU